

Shree Khangaonkar

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EDUCATION

University of Southern California

B.S. in Electrical and Computer Engineering, GPA: 3.77/4.00

Los Angeles, CA

Expected Graduation: 2028

EXPERIENCE

Sunnybotics

San Francisco, CA

Robotics Engineering Intern

May 2026 – Present

- Designed solar panel anomaly detector, and integrating it onto the TATACOABOT JC600, a **solar panel cleaning robot**, enabling georeferenced image capture during autonomous field missions.
- Developed the end-to-end perception data pipeline across camera/compute selection, GPS/odometry metadata tagging, storage/transmission, and integration with Sunnybotics' analysis workflow.
- Validated and deployed hardware on the robot through bench testing, provisional mounting, cable routing, vibration/glare mitigation, electrical debugging, and field-readiness checks.

Terra Labs: Cove

Los Angeles, CA

Electrical Lead

Nov 2025 – Present

- Led electrical integration for a **7-DOF** autonomous robotic arm built around a Teensy 4.1, Raspberry Pi 5, and CAN bus actuator network, achieving **controlled motion** across all 7 actuators on a **10-week** development timeline.
- Built the robot's wiring and power-distribution architecture, including actuator interfacing, CAN wiring, and E-Stop.
- Supported embedded communication and control architecture between ROS 2 software on the Raspberry Pi and firmware on the Teensy for joint-level actuation, feedback, and safety.
- Performed hardware bring-up, subsystem verification, and bench debugging to diagnose integration issues and improve system reliability while documenting failures, debugging steps, and design decisions.

PROJECTS

Custom Speaker Builder

March 2024 – Present

- Designed and built high-fidelity **Bluetooth** speakers from scratch, including enclosure modeling, analog crossover circuit design, and driver selection.
- Performed hands-on assembly with soldering, wiring, and acoustic/electrical testing using signal generators, oscilloscopes, and multimeters.
- Used iterative testing and debugging to refine circuit behavior and improve final system performance.

1 lb Battlebot

Aug 2025 – Dec 2025

- Designed, built, and used a 1 lb combat robot, integrating power and RF under competition constraints.
- Competed in a live event and finished with a **1–2** record, using post-match debugging and repair to improve system reliability.

Unconditional Audio Generator Model

September 2023 – March 2024

- Built a diffusion-based neural network to generate new audio clips from spectrogram inputs, achieving results comparable to state-of-the-art GAN-based approaches while reducing training time by **84%**.
- Implemented the model in PyTorch and trained it on one piece of music at a time, enabling low-data workflows that could run on consumer GPUs.
- Used Python for experimentation, data processing, and structured analysis of model behavior under limited compute constraints.

Experience Synesthesia

Project Lead / Chief Engineer

Sept 2025 – April 2026

- Built a sensory-blending art and engineering installation exploring the relationship between auditory and visual perception.
- Designed interactive sound-mixing podiums that allowed audiences to alter music in real time while generating evolving projected artwork.
- Integrated dynamic lighting and station interaction to create an immersive experience inspired by synesthesia.

SKILLS

Programming: Python, MATLAB, C, C++, Java, Swift, Arduino, HTML/CSS/JavaScript, LaTeX

Electrical / Systems: Embedded systems, hardware-software integration, CAN bus integration, analog/digital circuit design, KiCad, power distribution, schematic interpretation

Tools / Platforms: Linux command line, PowerShell, Teensy 4.1, Raspberry Pi 5, Arduino Nano/Uno, oscilloscopes, DMM, multimeters, signal generators, ROS 2, PyTorch

Strengths: Multidisciplinary teamwork, analytical problem solving, hands-on troubleshooting, clear communication, fast learner